



School of Electrical & Electronic Engineering

EE6108 Computer Networks

Academic Year 2025-2026

L6108

Enterprise Switched LAN Design and Implementation

Communication Research Laboratory (S2-B4c-17)

Dress Code in the Laboratory

- Work shirt that covers the upper torso and arms.
- Lower body clothing that covers the entire leg.
- Closed-toe shoes that cover the top of the foot.

Please access this link:

<http://tiny.cc/f6f5001> or scan the QR code below to access EEE CA Guidelines.

Request for MAKE-UP LAB SESSION must be submitted only after the online leave application form has been submitted.



Laboratory Manual

**NANYANG TECHNOLOGICAL UNIVERSITY
SCHOOL OF ELECTRICAL AND ELECTRONIC ENGINEERING**

ACADEMIC YEAR 2025/26

L6108 LABORATORY MANUAL

Enterprise Switched LAN Design and Implementation

**Communication Lab
(S2 – B4c – 17)**

Students are expected to read through all the theoretical parts of this manual before attending this laboratory session

NAME : _____

GROUP : _____

DATE : _____

INSTRUCTION

1. This 1.5 tadays laboratory session consists of three main parts:

Part I: IP Networking Fundamentals including Subnetting, Routing and Switching

Part II: Spanning Tree Protocol (STP) and Layer 2 Loop Prevention

Part III: VLAN Configuration, EtherChannel, and Inter-VLAN Routing

2. Enter your experimental results only in the space provided in this manual
3. Students may ask the lecturer-in-charge questions concerning the theory, laboratory equipment and experiment procedures but not questions set out in this manual.
4. Keep your answers short so that they can be written within the space provided in this manual.
5. Make sure you understand what you are doing and know how to operate the instruments. If you don't, please ask.

Part 1. IP Networking Fundamentals

1. Objectives

- To familiarize with Cisco Packet Tracer software interface and build a physical network topology independently.
- To perform IP Subnetting using standard formulas to divide a Class C network into multiple functional subnets.
- To study and understand the principles of Layer 2 Switching and Multi-link Physical Routing to enable cross-subnet communication.
- To configure DHCP (Dynamic Host Configuration Protocol) and utilize network troubleshooting skills to resolve broadcast domain conflicts.

2. Preparation

- Cisco Packet Tracer (Latest Version)
- 1 switch (Like 2960).
- 3 PC.
- 1 router (Like 2911).

3. Experiment 1 -- Topology Setup

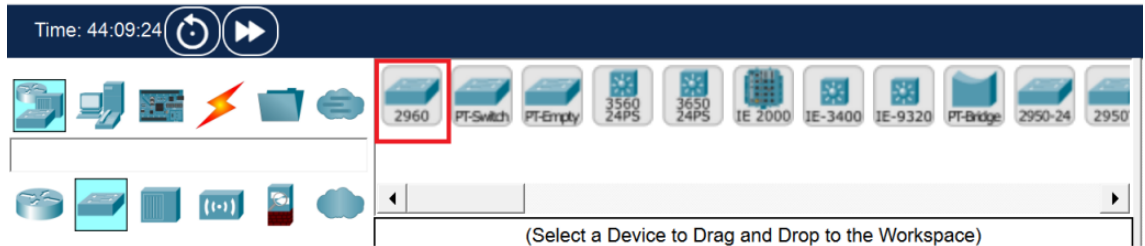
a. **Build the network topology in Packet Tracer**

Please complete the physical topology connection according to the instructions below:

- i. Open Cisco Packet Tracer.
- ii. In the lower-left corner of the Packet Tracer window, locate the Device-Type Selection Box.
- iii. Click End Devices, drag PC into the workspace, and place 3 PCs. Rename them as: PC0, PC1, and PC2.



- iv. Click Network Devices, then select Switches. Drag one 2960 Switch into the workspace.

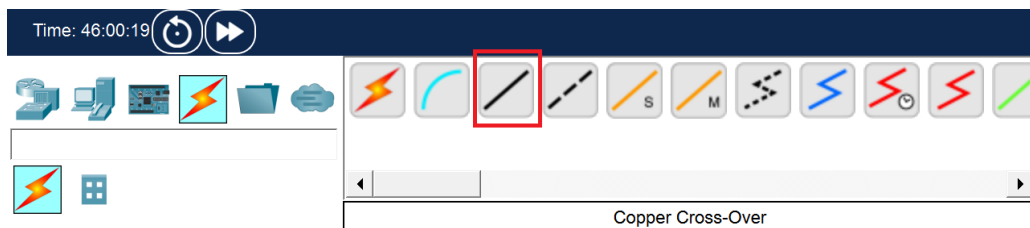


- v. Click Network Devices, then select Routers. Drag one 2911 Router (which provides 3 GigabitEthernet interfaces) and place it above the switch.



b. Cabling the Network

- i. Click Connections (the lightning icon) and select Copper Straight-Through cable.



- ii. Connect the devices according to the following tables to establish the 3 independent physical links for the departments:

Device	Interface	Connected Device	Interface
PC0	FastEthernet0	Switch0	FastEthernet0/1
PC1			FastEthernet0/2
PC2			FastEthernet0/3

Switch0	FastEthernet0/4	Router0	GigabitEthernet0/0
	FastEthernet0/5		GigabitEthernet0/1
	FastEthernet0/6		GigabitEthernet0/2

Note: The link lights on the router interfaces will initially appear as red triangles because the ports are administratively down by default. They will be activated in Experiment 3.

4. Experiment 2 — IP Subnetting

In this experiment, the base network address is given as 192.168.10.0/24. You are required to design a network for 3 departments, each requiring approximately 20 hosts. Calculate the required host bits(n) using the formula: $2^n - 2 \geq \text{required hosts}$.

a. **Calculation & Thinking**

- i. In the host calculation formula $2^n - 2 \geq \text{required hosts}$, why must we subtract 2 from the total permutations?

- ii. Based on the requirement of 20 hosts per department, calculate the required host bits (n) and derive the new subnet mask (please provide both CIDR notation and decimal format).

- iii. Taking the Marketing department as an example, if its starting network address is 192.168.10.32, what is its "Broadcast Address" based on the subnet size calculated in Q2? Can PC1 in the Marketing department configure this broadcast address on its own network interface?

b. Complete the IP Allocation Table:

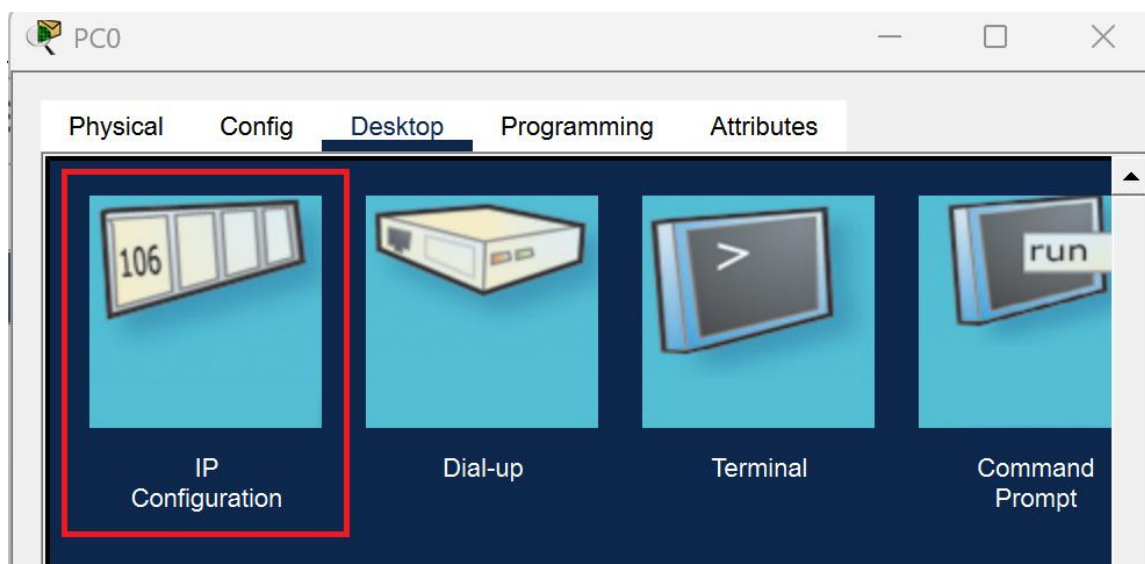
Department	Network Address	Usable IP Range	Subnet Mask	Default Gateway
Accounting (PC0)	192.168.10.0	192.168.10.1-.30		192.168.10.1
Marketing (PC1)	192.168.10.32			192.168.10.33
Engineering (PC2)	192.168.10.64			192.168.10.65

5. Experiment 3 — Switching & Routing Implementation

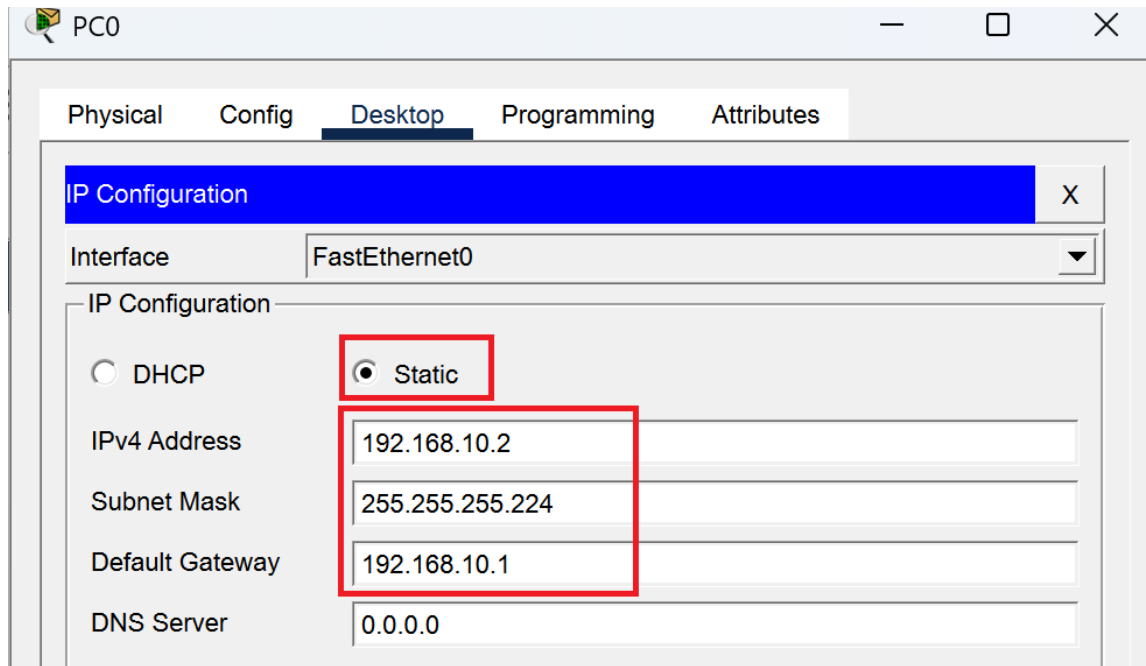
In this experiment, we will configure the end devices and the router to enable inter-subnet communication.

a. Configure PC IP address

- i. Click on PC0, navigate to the Desktop tab, and then click on IP Configuration.



- ii. Select Static. Set the IPv4 Address (e.g., 192.168.10.2), Subnet Mask (255.255.255.224), and Default Gateway (192.168.10.1) based on your calculation.



Note: When you click on this box, the software will automatically fill in 255.255.255.0.

Please make sure to manually delete the last 0 and change it to 224, making it 255.255.255.224. (This is the result of your Experiment 2 !)

- iii. Repeat this operation to configure PC1 and PC2 with their respective subnet data.

b. Configure the Layer 2 Switch

No configuration is required for the 2960 Switch. By default, all 24 interfaces are in the same broadcast domain (VLAN 1). It will automatically build its MAC address table to perform Layer 2 forwarding.

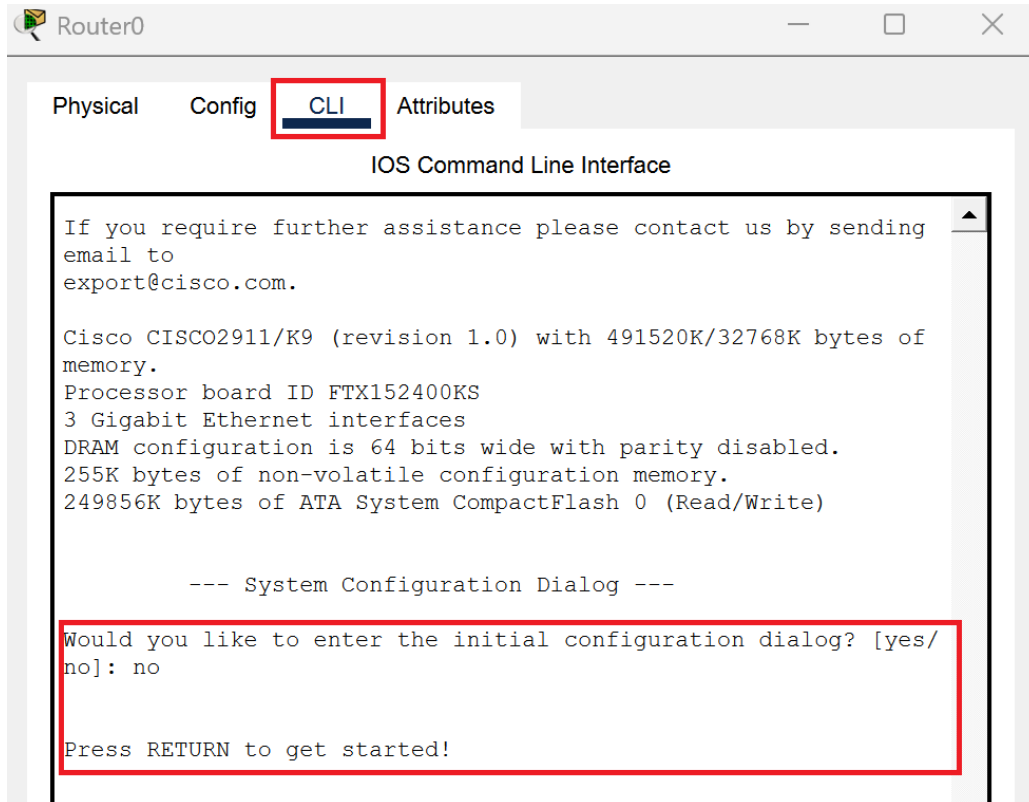
c. Configure the Layer 3 Router

- i. Click on PC0, navigate to the Desktop tab, and then click on IP Configuration.

Click on Router0, open the CLI tab.

If you see the prompt ' Would you like to enter the initial configuration dialog?

[yes/no]:', type ' no' and press Enter.



- ii. Press Enter a few times until the Router> prompt appears.

Enter the following commands to activate the interfaces and set the gateways:

```
Router> enable
```

```
Router# configure terminal
```

```
! --- Configure Accounting Gateway (G0/0) ---
```

```
Router(config)# interface gigabitEthernet 0/0
```

```
Router(config-if)# ip address 192.168.10.1 255.255.255.224
```

```
Router(config-if)# no shutdown
```

```
Router(config-if)# exit
```

```
! (Repeat the same logic to configure G0/1 for Marketing and G0/2 for Engineering)
```

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.224
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to
up

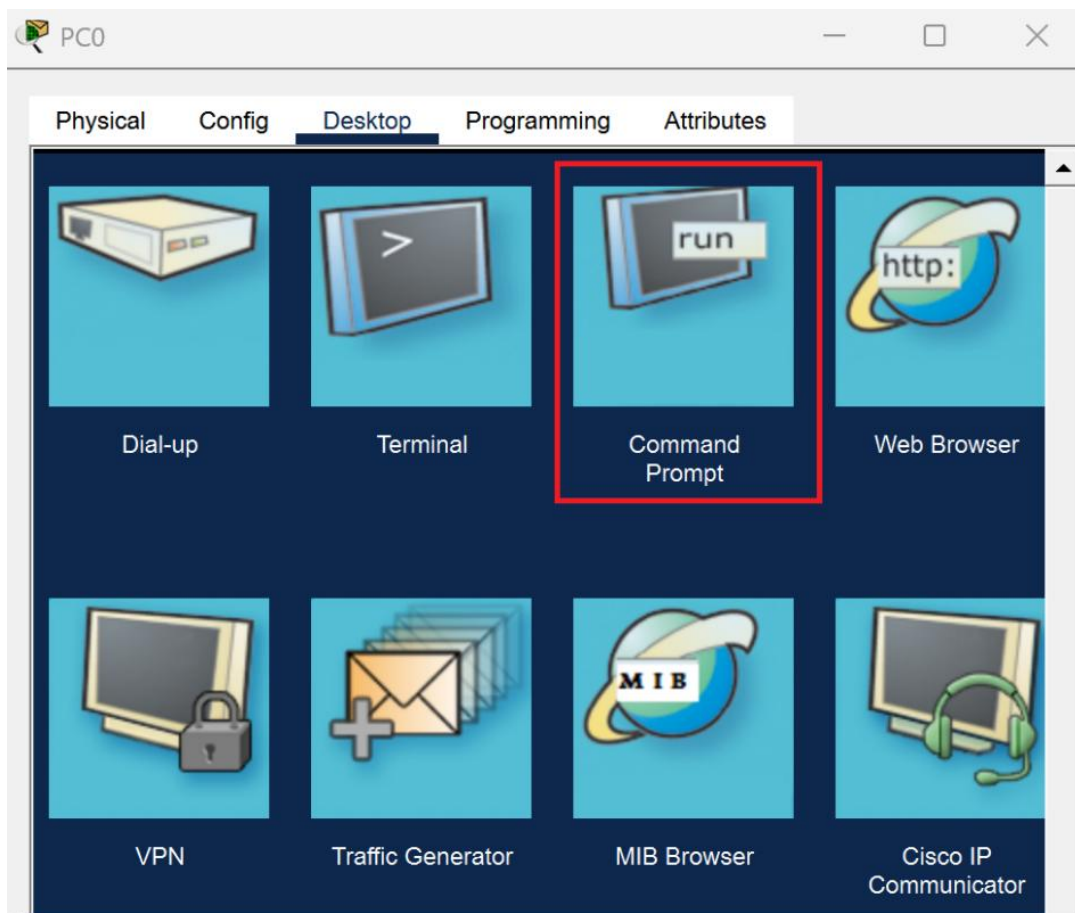
%LINEPROTO-5-UPDOWN: Line protocol on Interface
GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#
```

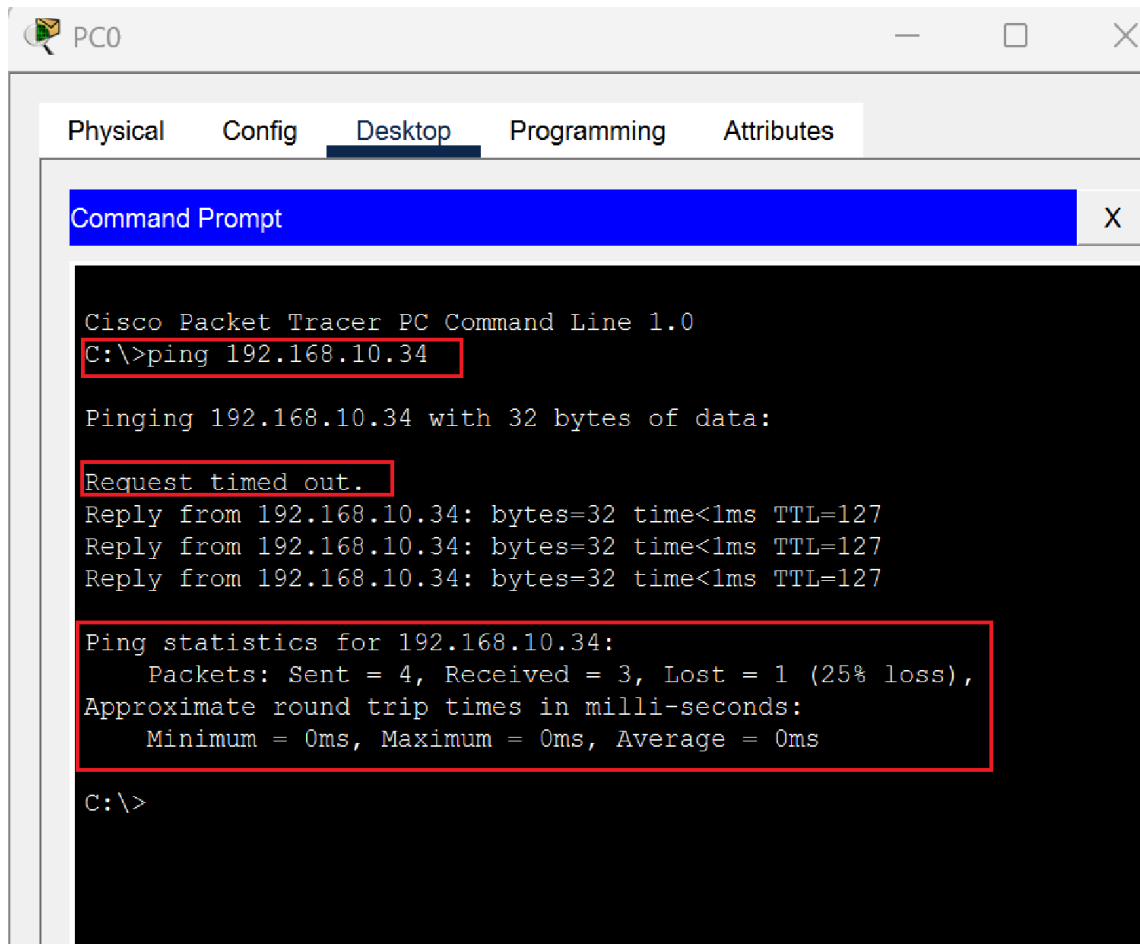
Copy Paste

d. Connectivity Verification

- i. Open the Command Prompt on PC0.



- ii. Enter the command 'ping 192.168.10.34' to test reachability to PC1.



The screenshot shows a Cisco Packet Tracer PC Command Line window for PC0. The window has tabs for Physical, Config, Desktop, Programming, and Attributes. The Desktop tab is active, displaying a Command Prompt window. The Command Prompt shows the following text:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.34

Pinging 192.168.10.34 with 32 bytes of data:

Request timed out.
Reply from 192.168.10.34: bytes=32 time<1ms TTL=127
Reply from 192.168.10.34: bytes=32 time<1ms TTL=127
Reply from 192.168.10.34: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.34:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Note: The first one or two requests may display "Request timed out" due to the ARP resolution process. This is a normal phenomenon. A subsequent "Reply from..." indicates successful routing.

6. Experiment 4 — DHCP & Broadcast Chaos

To improve efficiency, we will configure the router to act as a DHCP server. However, implementing this in a pure physical multi-link topology will trigger a network conflict.

a. **Configure DHCP Pools**

Enter the following commands in the Router CLI:

```
Router# configure terminal
```

```
! Exclude gateway IP addresses
```

```
Router(config)# ip dhcp excluded-address 192.168.10.1
```

```
Router(config)# ip dhcp excluded-address 192.168.10.33
```

```
Router(config)# ip dhcp excluded-address 192.168.10.65
```

! Create Accounting DHCP Pool

```
Router(config)# ip dhcp pool Pool_Acc
```

```
Router(dhcp-config)# network 192.168.10.0 255.255.255.224
```

```
Router(dhcp-config)# default-router 192.168.10.1
```

```
Router(dhcp-config)# exit
```

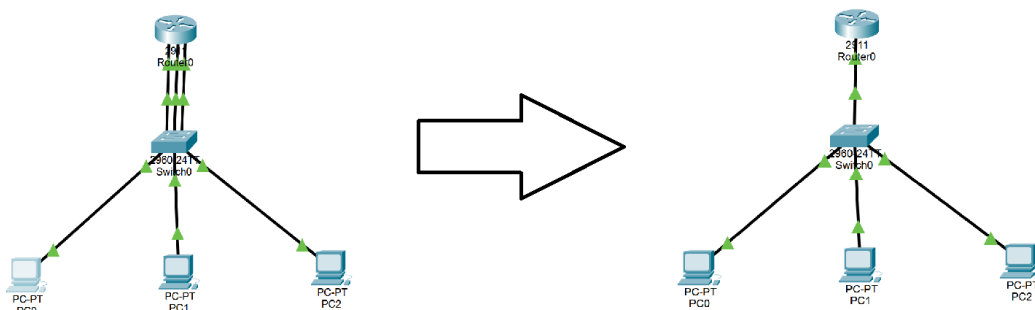
! (Proceed to create Pool_Mar and Pool_Eng...)

b. Observe the Bug & Physical Verification

- i. On PC0, switch the IP Configuration mode from Static to DHCP. Observe the obtained IP address and Gateway. (You will notice PC0 receives an incorrect IP and Gateway pairing).

IP Configuration	
☒ DHCP	☐ Static
DHCP request successful.	
IPv4 Address	192.168.10.67
Subnet Mask	255.255.255.224
Default Gateway	192.168.10.1
DNS Server	0.0.0.0

- ii. Select the Delete tool in Packet Tracer. Remove two of the physical links between the switch and the router, keeping only the G0/0 link.



- iii. Switch PC0 back to Static and then to DHCP again. Observe the successful allocation.

Interface	FastEthernet0
IP Configuration	
☒ DHCP	☐ Static DHCP request successful.
IPv4 Address	192.168.10.6
Subnet Mask	255.255.255.224
Default Gateway	192.168.10.1
DNS Server	0.0.0.0

c. Practice & Thinking Question

Based on the concept of "Broadcast Domain", explain why the DHCP allocation fails when 3 links are connected, but succeeds when 2 links are removed?

Note: After completing this experiment, reconnect the 2 deleted links using Copper Straight-Through cables and set all PCs back to Static IP mode for the next phase.

7. Transition to Next Parts

By removing the physical cables in Experiment 4, we temporarily solved the DHCP broadcast conflict, but introduced critical engineering challenges:

a. Reliability Crisis:

Relying on a single switch creates a single point of failure. Introducing redundant switches will cause Layer 2 loops. (To be resolved in Part 2. Spanning Tree Protocol - STP).

b. Scalability Crisis:

Unplugging cables disables other departments. Managing a network without logical boundaries is unsustainable. (To be resolved in Part 3. Virtual LAN - VLAN).

Part 2. Spanning Tree Protocol (STP) and Layer 2 Loop Prevention

1. Objectives

- To study and understand the basic principles of Spanning Tree Protocol (STP).
- To build a switched network topology and configure IP connectivity between hosts.
- To introduce redundant links between switches and observe how STP prevents Layer 2 loops.
- To analyze STP information and port roles using CLI commands.
- To understand the Root Bridge election process and influence it by modifying bridge priority.
- To observe how STP recalculates the topology and activates a backup path when a link failure occurs.

2. Preparation

- Cisco Packet Tracer (Lastest Version)
- Lab Manual with Student instruction and Packet tracer file used in the workshop

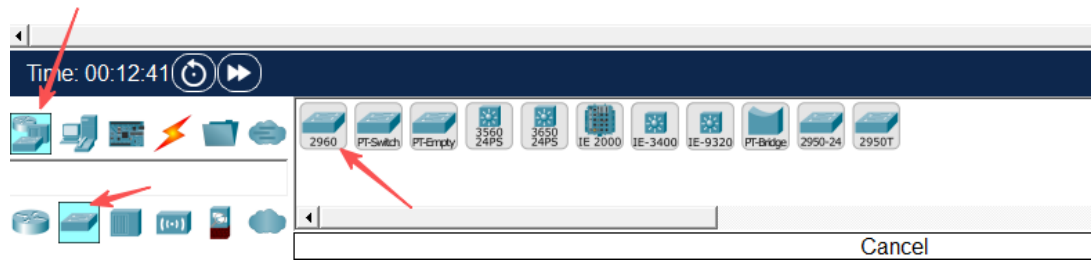
3. Experiment 1 Topology setup

Please complete the terminal device configuration according to the instructions below:

a. Build the network topology in Packet Tracer

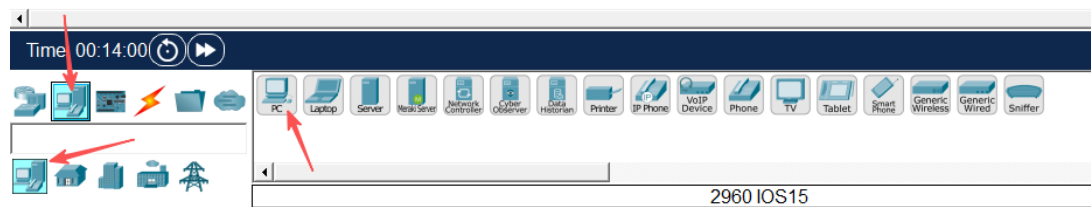
- iv. Open Cisco Packet Tracer.
- v. In the lower-left corner of the Packet Tracer window, locate the Device-Type Selection Box.
- vi. Click Network Devices, then select Switches.
- vii. From the device list, drag 2960 Switch into the workspace.

Place two switches and rename them as: Switch0, Switch1



viii. In the Device-Type Selection Box, click End Devices.

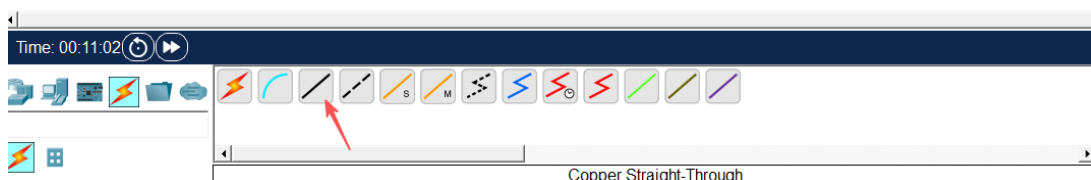
ix. Drag PC into the workspace and place six PCs. Rename them from PC0 to PC5.



x. Click Connections (the lightning icon).

xi. Select Copper Straight-Through cable and connect the PCs to the switches according to the following table:

Device	Interface	Connected Device	Interface
PC0	FastEthernet0	Switch0	FastEthernet0/1
PC1			FastEthernet0/2
PC2			FastEthernet0/3
PC3		Switch1	FastEthernet0/1
PC4			FastEthernet0/2
PC5			FastEthernet0/3

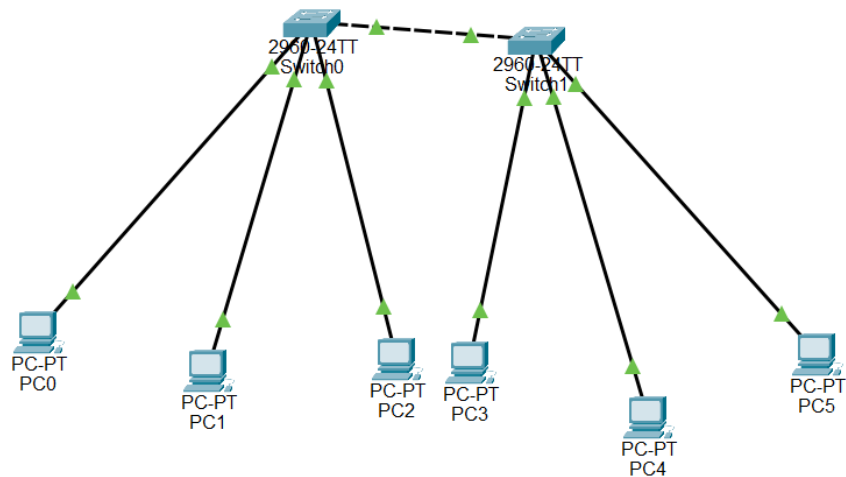


xii. Use Copper cross-over cable to connect the two switches:

Device	Interface	Connected Device	Interface
--------	-----------	------------------	-----------

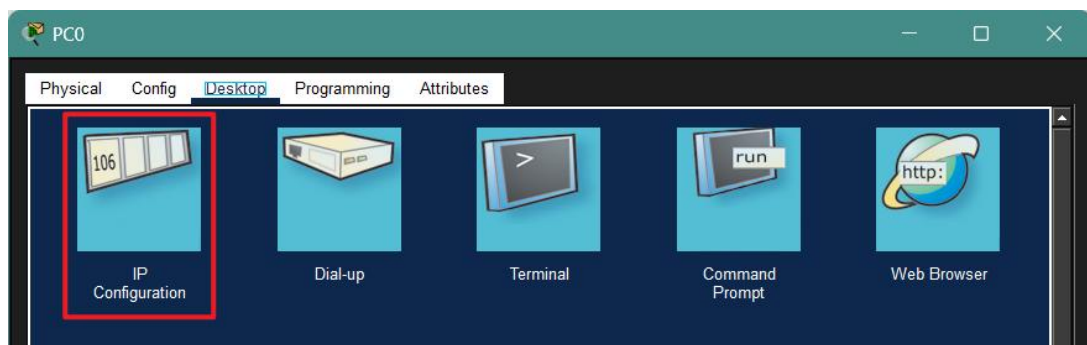
Switch0	FastEthernet0/4	Switch1	FastEthernet0/4
---------	-----------------	---------	-----------------

After completing the connections, ensure that all link lights turn green, indicating that the connections are active.

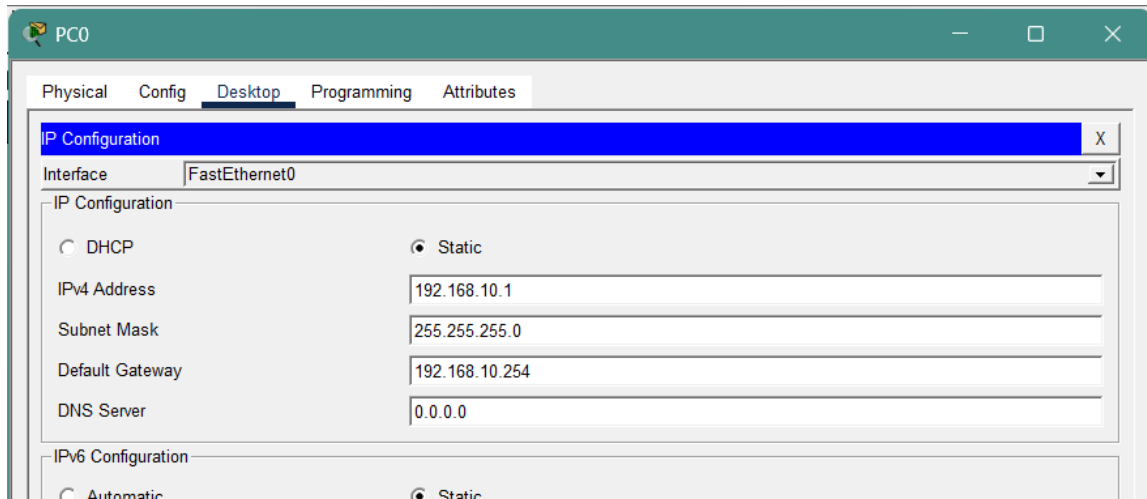


b. Configure PC IP address

- i. Click on PC0, navigate to the Desktop tab, and then click on IP Configuration.



- ii. Set the following parameters:
 IPv4 Address: 192.168.10.1
 Subnet Mask: 255.255.255.0
 Default Gateway: 192.168.10.254



- iii. Repeat this operation to configure PC1-PC5 with IPv4 addresses ranging from 192.168.10.2 to 192.168.10.6.

PC Configuration Reference:

PC No.	IPv4 Address	Subnet Mask	Default Gateway
PC0	192.168.10.1	255.255.255.0	192.168.10.254
PC1	192.168.10.2		
.....			
PC5	192.168.10.6		

Note: You can choose not to follow the table setup. But do configure all six PCs on the same network segment.

4. Experiment 2 Introduce Loop and Observe STP Blocking

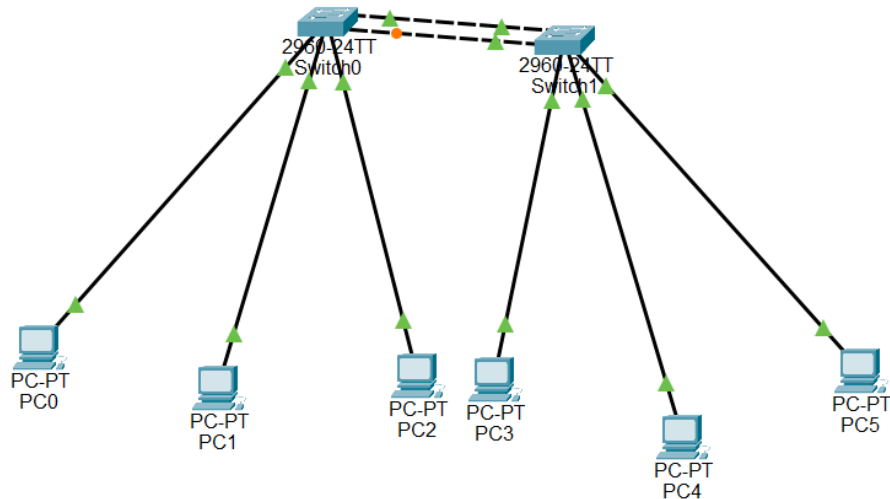
In this experiment, an additional physical link will be introduced between the two switches to create a Layer 2 loop.

a. Add an additional link between the switches

- In Packet Tracer, click Connections (the lightning icon).
- Select Copper cross-over cable.
- Connect an additional link between the two switches using the following interfaces:

Device	Interface	Connected Device	Interface
--------	-----------	------------------	-----------

Switcho	FastEthernet0/5	Switcho	FastEthernet0/5
---------	-----------------	---------	-----------------



After completing the connection, there will now be two physical links between the two switches. Since there are now two parallel links between the switches, a Layer 2 loop is created in the network.

b. Observe the switch port status

When a loop appears in a Layer 2 network, the switch automatically uses the Spanning Tree Protocol (STP) to prevent network loops.

- i. Observe the port status indicators on the switch links in Packet Tracer.
- ii. Among the two links connecting the switches, one of the ports will change to Orange/Green/Red
- iii. This indicates that STP has automatically Blocked/Opened one of the redundant links to prevent a network loop.

c. Verify STP information using CLI

- i. Click Switch1 and open the CLI tab.
- ii. Enter the following commands:

Switch> **enable**

Switch# **show spanning-tree**

- iii. From the output, observe the following information:

```
Switch>Enable
Switch#show spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     000C.CFBE.13C5
             This bridge is the root
             Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
             Address     000C.CFBE.13C5
             Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface                Role Sts Cost          Prio.Nbr Type
-----
Fa0/2                    Desg FWD 19           128.2    P2p
Fa0/1                    Desg FWD 19           128.1    P2p
Fa0/3                    Desg FWD 19           128.3    P2p
Fa0/4                    Desg FWD 19           128.4    P2p
Fa0/5                    Desg FWD 19           128.5    P2p
```

These values show how STP determines the forwarding path and which port should be blocked.

d. Practice & Thinking Questions

- i. Why does one of the switch ports enter the Blocking state after adding a second link?

- ii. What problems might occur if the network operates without STP?(at least two points)

- iii. What is the main purpose of Spanning Tree Protocol in a Layer 2 network?

- iv. Complete the STP Information Table of Switch0:

Item	Value
Root Bridge Priority	
Root Bridge MAC Address	
Root Path Cost	
Root Port	
Designated Ports	
Blocking Port	

5. Experiment 3 Root Bridge Election

In this experiment, we will investigate how Spanning Tree Protocol (STP) elects the Root Bridge in a network and observe how changing the bridge priority can influence the election result.

a. Observe the current root bridge

- i. Click on Switch0 and open the CLI tab.
- ii. Enter the following commands:

```
Switch> enable
```

```
Switch# show spanning-tree
```
- iii. Repeat the same command on Switch1 and find the root bridge.

b. Change & Verify the bridge priority

STP selects the Root Bridge based on the Bridge ID, which is composed of:

Bridge ID = Priority + MAC Address

A switch with the lowest Bridge ID will become the Root Bridge.

- i. On Switch0, enter the following commands:

```
Switch> enable
```

```
Switch# configure terminal
```

```
Switch(config)# spanning-tree vlan 1 priority 4096
```

This command lowers the bridge priority of Switch0, increasing its chance of becoming the Root Bridge.

- ii. On Switch0, run the following command again:

Switch# **show spanning-tree**

- iii. Check whether the output now shows:

This bridge is the root

- iv. Run the same command on Switch1 and observe how the Root Bridge information changes.

c. Practice & Thinking Questions

- i. What factor determines which switch becomes the Root Bridge?

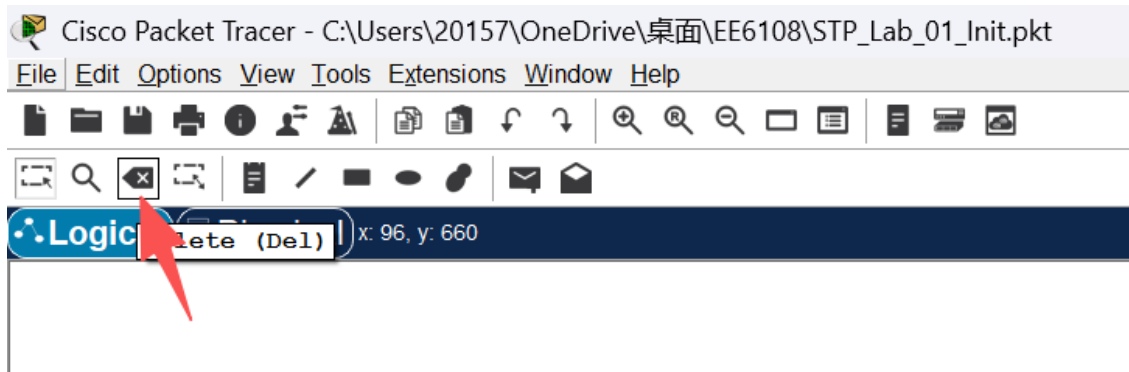
- ii. After changing the Root Bridge, does the Blocking port change? Explain why.

6. Experiment 4 Link Failure Recovery

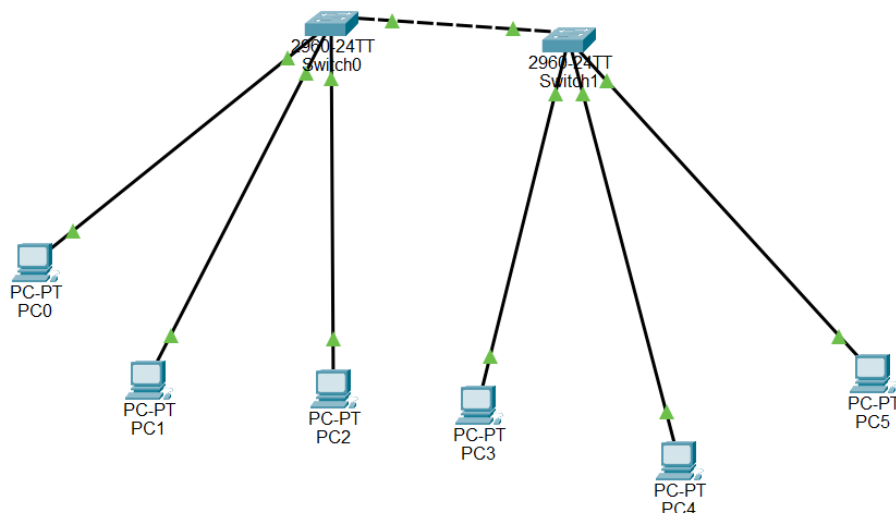
In this experiment, we will observe how Spanning Tree Protocol (STP) responds to a link failure and automatically activates a backup path.

a. Disconnect the active link and observe STP reconvergence

- i. In Packet Tracer, select the Delete tool.



- ii. Identify the port that is currently in the Blocking state.
- iii. Remove the currently active link between the two switches. This link is the one whose ports are in the Forwarding state (Green).
- iv. Identify the port that was previously in the Blocking state. After a short period of time, the port state will change from Blocking (Orange) to Forwarding (Green).



- v. Click Switch0 and open the CLI tab.
- vi. Enter the following command to view the STP information:
 Switch> **enable**
 Switch# **show spanning-tree**
- vii. Observe the changes in the port roles and port states.

b. Practice & Thinking Questions

- i. What happens to the previously blocked port after the active link fails?

ii. What is the advantage of having redundant links in a network?

Part 3. Virtual LAN(VLAN) Management, Link Aggregation, and Inter-VLAN Routing

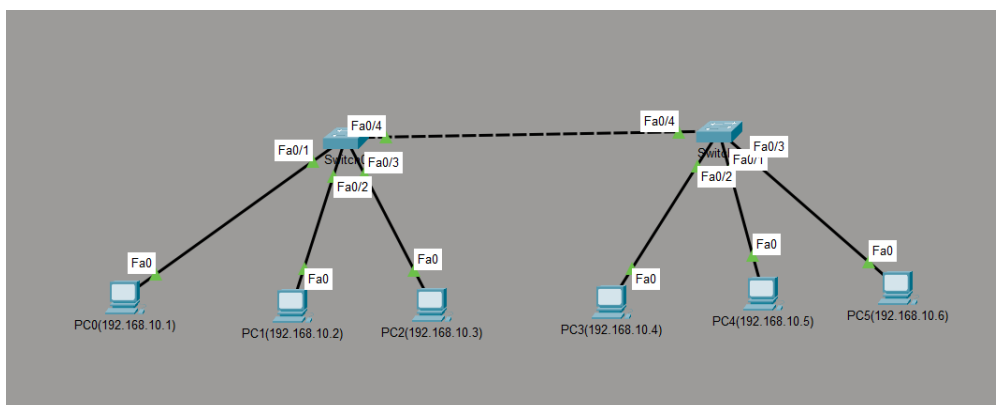
1. Objectives

- To study and practice the principles of VLAN.
- Create multiple VLANs and assign corresponding ports.
- Create a trunk line between 2 switches for paired VLANs.
- By implementing the **port security** understands Layer 2 Security.
- Understand the implementation of **EtherChannel**.
- implement **Single-Arm Routing** for communication between different IP subnets.

2. Preparation

- 2 switches (Like 2960).
- 6 PC.
- Each switch is connected to 3 PCs.
- Connect the 2 switches.

3. Description of laboratory experiment set-up



4. Experiment 1 - Basic Configuration

a. Create VLANs and name them

Repeat the following commands on both switches (Switch0 and Switch1):

```

Switch> enable

Switch# configure terminal

Switch(config)# vlan 10

Switch(config-vlan)# name EEE

Switch(config-vlan)# vlan 20

Switch(config-vlan)# name NIE

Switch(config-vlan)# vlan 30

Switch(config-vlan)# name CCDS

Switch(config-vlan)# exit

```

```

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to down
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up

Switch>
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name EEE
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name NIE
Switch(config-vlan)#vlan 30
Switch(config-vlan)#name CCDS
Switch(config-vlan)#exit
Switch(config)#

```

Or you can choose to change the config on both switches (Switch0 and Switch1):

VLAN Configuration

VLAN Number:

VLAN Name:

VLAN No	VLAN Name
1	default
10	EEE
20	NIE
30	CCDS
1002	fddi-default
1003	token-ring-default
1004	fddinet-default
1005	trnet-default

b. Assign access port

Execute on Switch0 and Switch1 respectively:

- i. Assign PC0 / PC3 to VLAN 10 (interface Fa0/1):

```
Switch(config)# interface FastEthernet0/1  
Switch(config-if)# switchport mode access  
Switch(config-if)# switchport access vlan 10
```

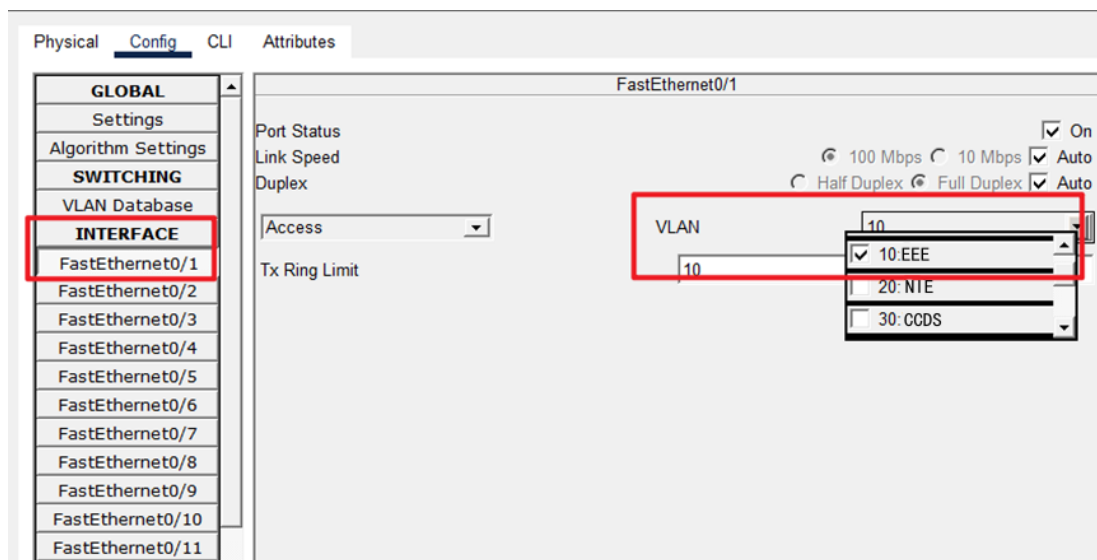
- ii. Assign PC1 / PC4 to VLAN 20 (interface Fa0/2):

```
Switch(config)# interface FastEthernet0/2  
Switch(config-if)# switchport mode access  
Switch(config-if)# switchport access vlan 20
```

- iii. Assign PC2 / PC5 to VLAN 30 (interface Fa0/3):

```
Switch(config)# interface FastEthernet0/3  
Switch(config-if)# switchport mode access  
Switch(config-if)# switchport access vlan 30
```

Or change according FastEthernet config:

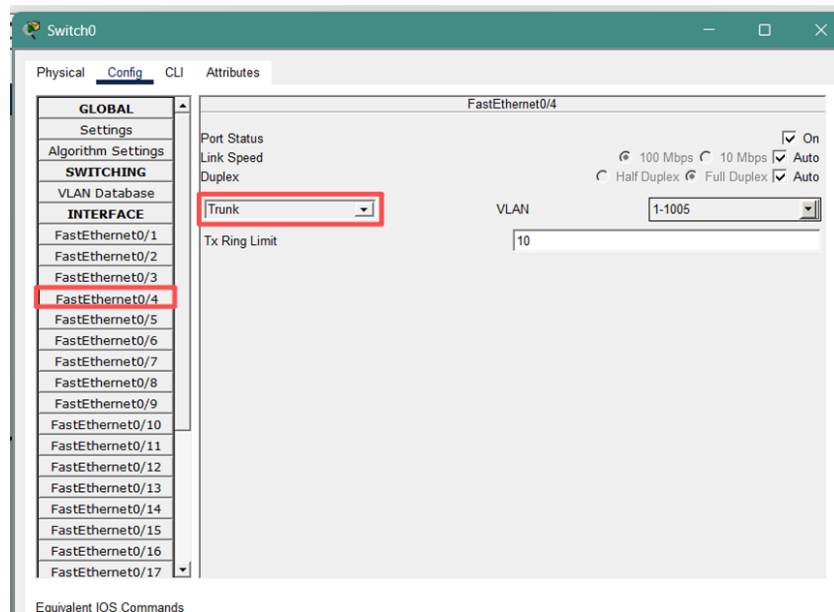


c. Configure trunk

Execute on Switch0 and Switch1 respectively:

```
Switch(config)# interface FastEthernet0/4  
Switch(config-if)# switchport mode trunk
```

Or change according ethernet config:



Note: After executing this command, the link lights in Packet Tracer may temporarily turn red and then green, indicating that the **Spanning Tree Protocol (STP)** is converging again.

d. View VLAN & trunk & interfaces information

Enter “enable” at the “Switch> ” (At Switch’s CLI interface),

- To view VLAN allocation, input: “**show vlan brief**”:
- Check whether Trunk is enabled: “**show interfaces trunk**”
- Check interface status: “**show ip interface brief**”

e. Practice & Thinking Questions

- Test 1 (VLAN Cross-Switch): Please ping the VLAN 10 host of Switch 1 from the VLAN 10 host of Switch 0. What is the result? What does this indicate about the role of the Trunk link?

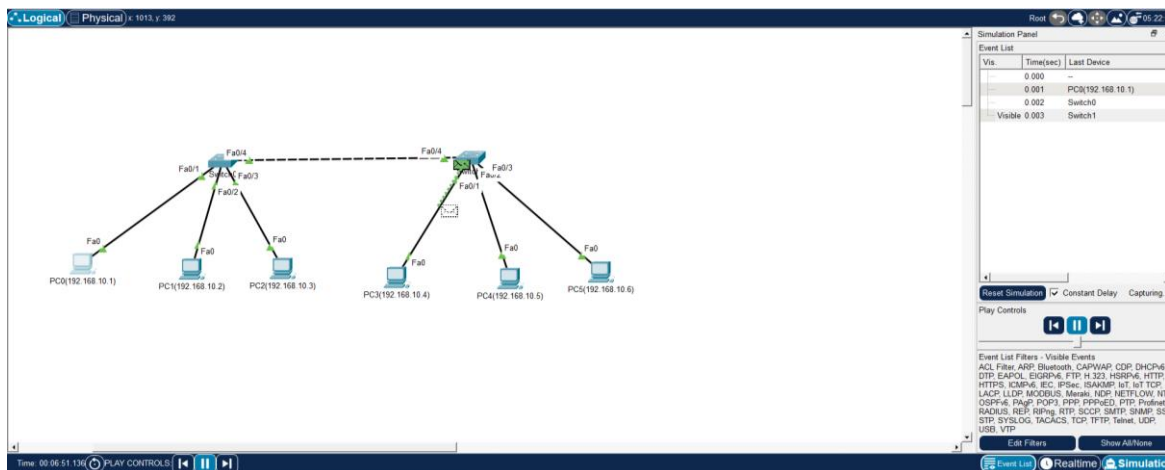
- Test 2 (Communication across Different VLANs): Please ping the host on VLAN 20 of

Switch 0 from the host on VLAN 10 of Switch 0. What is the result? Explain why this result occurs, taking into account the MAC address table (show mac address-table).

If you want to prove that "only VLAN 10 can receive", you can use the Simulation mode of Packet Tracer:

Click the clock icon in the lower right corner to switch to Simulation mode.

In the "Edit Filters" section, only check the ARP option.



Let PC0 ping PC3.

Click the "Play" button, and you will see the green packet (ARP) flying out like an envelope.

Observation: You will notice that the envelope only flies to ports within the same VLAN, while a red "X" appears above other VLAN ports, indicating that they have been blocked.

- iii. Test 3: Change one end of the Trunk link between two switches back to Access mode, observe the changes in network communication, and consider what consequences would arise from a mismatch in negotiation between the two ends.

-
-
- iv. Q1(Broadcast Domain Isolation): When no VLAN is configured, if a PC sends an ARP broadcast request, which devices in the network can receive it? After configuring VLANs 10, 20, and 30, if a PC within VLAN 10 sends an ARP broadcast, who else can receive it? Please briefly describe how VLAN improves network performance and security.
-
-
-
-

- v. Q2(The Essence of Trunk): If we do not configure a Trunk interface, but instead use three network cables to connect the VLAN 10, 20, and 30 interfaces of two switches respectively, can we achieve the same communication effect? If so, why do we still use Trunk technology?
-
-
-
-

- vi. Q3(No routing state): In the experiment, different VLANs cannot communicate with each other. If we want VLAN 10 and VLAN 20 to communicate with each other, what type of device do we need to introduce into the network topology? (To pave the way for the next stage of the single-arm router/layer 3 switch experiment)
-
-
-
-

Note: Congratulations on completing Experiment 1 !

The following 3 experiments are based on this experiment (Experiment 1). Please save

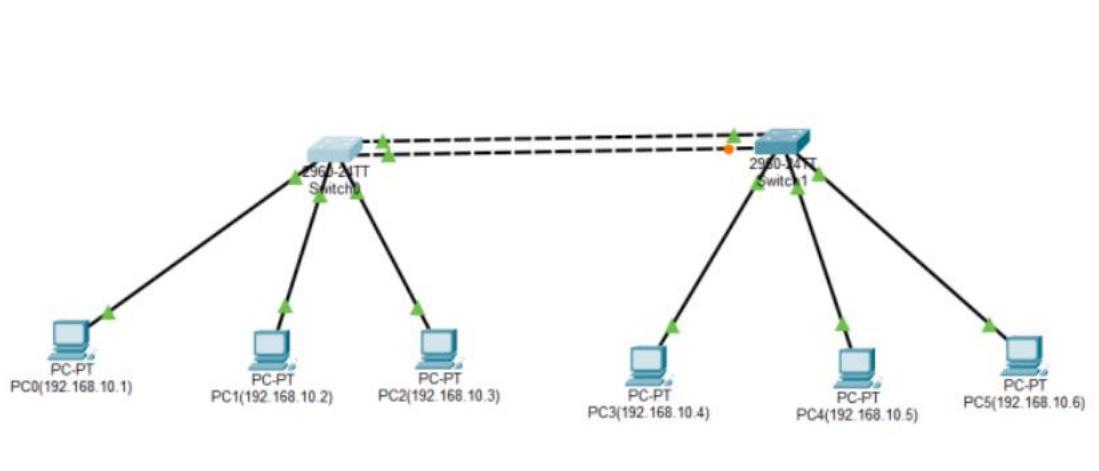
the .pkt file current as a template for subsequent experimental operations.

5. Experiment 2 -- EtherChannel and Spanning Tree Protocol (STP)

Observation

a. Upgrade Topology and Observe the initial state of STP

Connect an additional crossover cable between two 2960 switches. After connecting the cable without further configuration, one port on one of the links will turn orange (Blocking state).



Enter *show spanning-tree* in the privileged mode to observe the root bridge election results and specific port statuses (blocked ports are displayed as BLK)

```
Switch>enable
Switch#show spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
            Address    0090.2164.4740
            Cost        19
            Port        4(FastEthernet0/4)
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
            Address    00D0.58A4.7323
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
            Aging Time  20

Interface    Role Sts Cost      Prio.Nbr Type
-----
Fa0/5        Altn BLK 19      128.5    P2p
Fa0/4        Root FWD 19      128.4    P2p
VLAN0010
  Spanning tree enabled protocol ieee
  Root ID    Priority    32778
            Address    0090.2164.4740
            Cost        19
            Port        4(FastEthernet0/4)
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32778 (priority 32768 sys-id-ext 10)
  --More--
```

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b. Configure EtherChannel

- i. In the global configuration mode of Switch 0 and Switch 1, enter the following identical configuration commands:

enable

show spanning-tree

1.Enter the two connected physical interfaces simultaneously.

interface range fastEthernet 0/4-5

2.Bundle these two ports into Channel Group 1 and set them to active negotiation mode.

channel-group 1 mode active

exit

Enter the newly created logical interface Port-Channel 1.

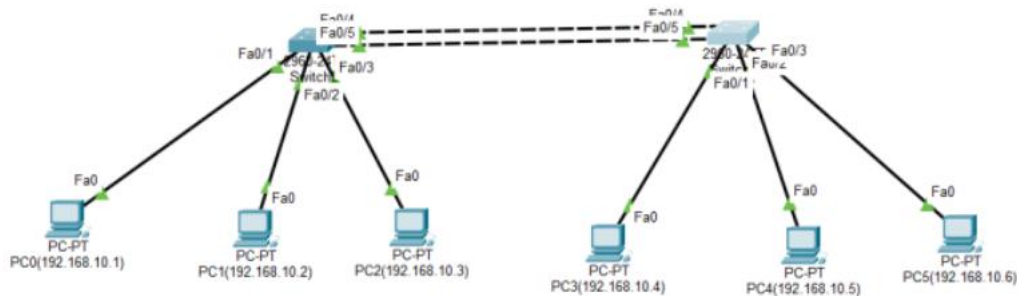
interface port-channel 1

#Configure this logical interface as Trunk mode to carry all VLANs.

switchport mode trunk

End

- ii. Use the Fast Forward Time feature in Packet Tracer to manually accelerate the convergence of STP and LACP (Link Aggregation Control Protocol). Once the network converges, all four port indicators between the switches turn green. Communication between PC1 and PC4 remains normal.



c. Verification Procedures

Execute the following three core commands in the privileged mode (Switch#) of Switch 0 or Switch 1:

- i. Check Aggregation Summary Status

Command: Switch# show etherchannel summary

```
Switch>enable
Switch#show etherchannel summary
Flags: D - down        P - in port-channel
       I - stand-alone s - suspended
       H - Hot-standby (LACP only)
       R - Layer3      S - Layer2
       U - in use      f - failed to allocate aggregator
       u - unsuitable for bundling
       w - waiting to be aggregated
       d - default port

Number of channel-groups in use: 1
Number of aggregators:          1

Group  Port-channel  Protocol    Ports
-----
1      Po1(SU)          LACP       Fa0/4(P) Fa0/5(P)
```

Check Group Status if Po1 is marked as (SU). S stands for Layer 2, and U stands for In Use, which signifies success. Check Ports Status if physical ports (e.g., Fa0/4, Fa0/5) are marked as (P), indicating they are successfully bundled into the Port-channel.

- ii. Check Spanning Tree Protocol (STP) Status

command: Switch# show spanning-tree

```
Switch#show spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0090.2164.4740
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     0090.2164.4740
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface    Role Sts Cost      Prio.Nbr Type
-----
Po1          Desg FWD 12        128.27  F2p

VLAN0010
  Spanning tree enabled protocol ieee
  Root ID    Priority    32778
             Address     0090.2164.4740
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32778  (priority 32768 sys-id-ext 10)
             Address     0090.2164.4740
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
--More--
```

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The original physical ports Fa0/4 and Fa0/5 should disappear from the list, replaced by the logical interface Po1 in the FWD (Forwarding) state.

- iii. Final Connectivity Test

Perform a continuous Ping test from PC0 (VLAN 10) to PC3 (VLAN 10). Ping responses are normal with 0% loss.

```
PC0(192.168.10.1)
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.4 -t
Invalid Command.
C:\>cls
Invalid Command.
C:\>ping 192.168.10.4
Pinging 192.168.10.4 with 32 bytes of data:
Reply from 192.168.10.4: bytes=32 time=1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time=1ms TTL=128
Ping statistics for 192.168.10.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>
```

- iv. Why did the orange port (blocked by STP) turn green after configuring EtherChannel?

7. Experiment 3 -- Port Security Configuration

a. Port security configuration command (CLI)

Please click on Switch0 to enter the CLI tab, and then enter the following commands in sequence:

Switch> **enable**

Switch# **configure terminal**

Switch(config)# **interface FastEthernet0/3**

The port must first be set to Access mode (Port Security does not support Trunk or dynamic ports)

Switch(config-if)# **switchport mode access**

Enable port security function

Switch(config-if)# **switchport port-security**

Limit the maximum number of MAC addresses allowed to be learned to 1

Switch(config-if)# **switchport port-security maximum 1**

Set to Sticky, permanently binding the currently connected PC's MAC address to the configuration

Switch(config-if)# **switchport port-security mac-address sticky**

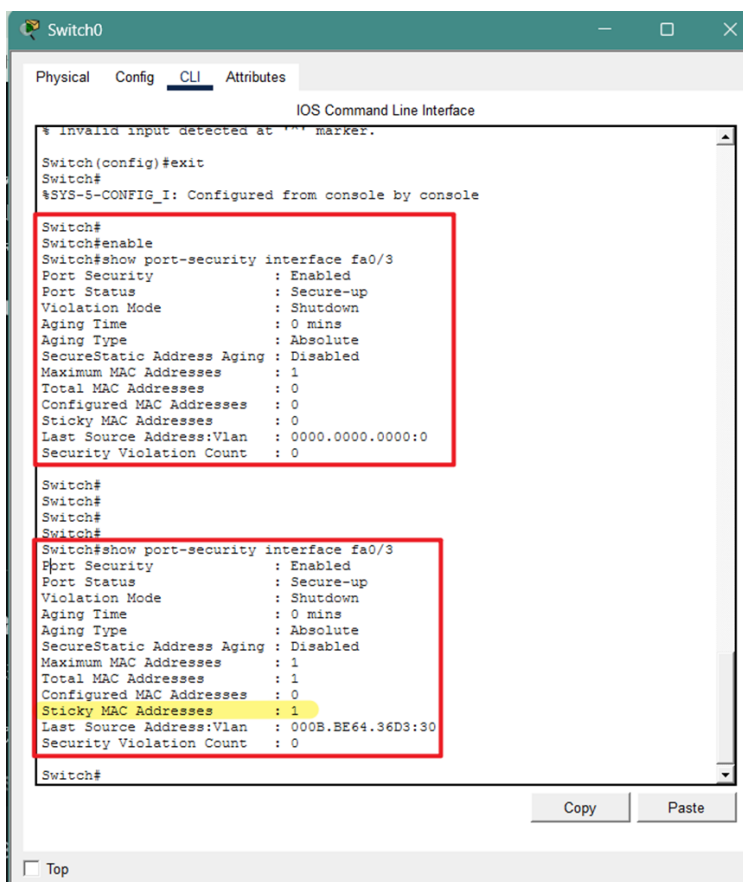
Set the violation triggering action (default is shutdown, which means closing the port)

Switch(config-if)# **switchport port-security violation shutdown**

b. Ensure that the switch "captures" the identity of legitimate hosts

Ping other PCs in the net. Making sure that the switch remember the MAC address.

You can confirm that the Sticky MAC Addresses has changed to 1 by entering "show port-security interface fa0/3" on the switch.



The screenshot shows a Cisco IOS Command Line Interface window titled "Switch0". The "CLI" tab is selected. The interface shows the following commands and output:

```
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#
Switch#enable
Switch#show port-security interface fa0/3
Port Security          : Enabled
Port Status            : Secure-up
Violation Mode         : Shutdown
Aging Time             : 0 mins
Aging Type             : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses  : 1
Total MAC Addresses    : 0
Configured MAC Addresses : 0
Sticky MAC Addresses   : 0
Last Source Address:Vlan : 0000.0000.0000:0
Security Violation Count : 0

Switch#
Switch#
Switch#
Switch#
Switch#show port-security interface fa0/3
Port Security          : Enabled
Port Status            : Secure-up
Violation Mode         : Shutdown
Aging Time             : 0 mins
Aging Type             : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses  : 1
Total MAC Addresses    : 1
Configured MAC Addresses : 0
Sticky MAC Addresses   : 1
Last Source Address:Vlan : 000B.BE64.36D3:30
Security Violation Count : 0

Switch#
```

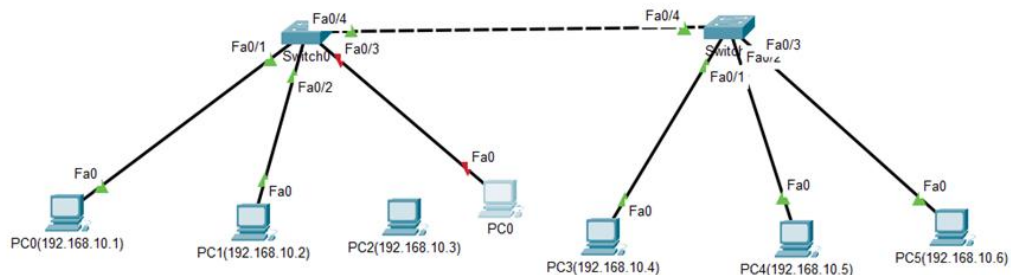
Two red boxes highlight the output of the "show port-security interface fa0/3" command. The first box shows the initial state where "Sticky MAC Addresses" is 0. The second box shows the state after a MAC address has been learned, where "Sticky MAC Addresses" is 1 and "Total MAC Addresses" is 1. The "Last Source Address:Vlan" is also shown as 000B.BE64.36D3:30.

c. Simulating "hacker" access

Disconnect: Use the delete tool (shortcut key Del) to remove the connection between Switch0 and the original PC2.

Connect a new device: Add a new laptop or PC and connect it to Fa0/3 of Switch0.

Trigger: When configuring the IP address for the hacker's PC, you will notice that the light on the Fa0/3 interface instantly changes from green to red.



At this time, you can view the following information by checking the CLI interface of the switch:

```
Switch0
Physical Config CLI Attributes
IOS Command Line Interface
Port Status : Secure-up
Violation Mode : Shutdown
Aging Time : 0 mins
Aging Type : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses : 1
Total MAC Addresses : 0
Configured MAC Addresses : 0
Sticky MAC Addresses : 0
Last Source Address:Vlan : 0000.0000.0000:0
Security Violation Count : 0

Switch#
Switch#
Switch#
Switch#
Switch#show port-security interface fa0/3
Port Security : Enabled
Port Status : Secure-up
Violation Mode : Shutdown
Aging Time : 0 mins
Aging Type : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses : 1
Total MAC Addresses : 1
Configured MAC Addresses : 0
Sticky MAC Addresses : 1
Last Source Address:Vlan : 000B.BE64.36D3:30
Security Violation Count : 0

Switch#
%LINK-3-UPDOWN: Interface FastEthernet0/3, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to down
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to administratively down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to down
%PM-4-ERR_DISABLE: psecure-violation error detected on Fa0/3, putting Fa0/3 in err-disable state.
%PORT_SECURITY-2-PSECURE_VIOLATION: Security violation occurred, caused by MAC address 0060.2FCA.EC7C on port FastEthernet0/3.
```

d. Phenomenon observation and recovery

When a port is disabled due to violations, it enters a special state known as "err-disabled".

i. View violation records

Enter the follow commands at Switch's CLI tab(Switch#):

Switch# show port-security interface FastEthernet0/3

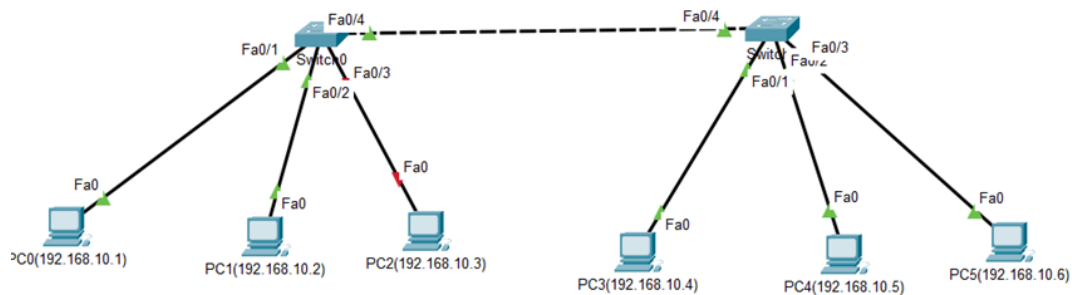
You will see in the output:

Port Status: Security-shutdown

Security Violation Count: 1

Last Source Address: [The MAC address of the hacker's PC will be displayed here]

ii. Restore interface



First, reconnect the originally legal PC.

Enter the following command at switch's CLI tab:

Switch# configure terminal

Switch(config)# interface FastEthernet0/3

Switch(config-if)# shutdown

Switch(config-if)# no shutdown

It can be observed that the original PC has restored normal connection.

Question: If you directly enter "not shutdown" instead of "shutdown" first, will the original PC be able to restore normal connection? What is the status of the interface?

Why?

e. Auto Recovery

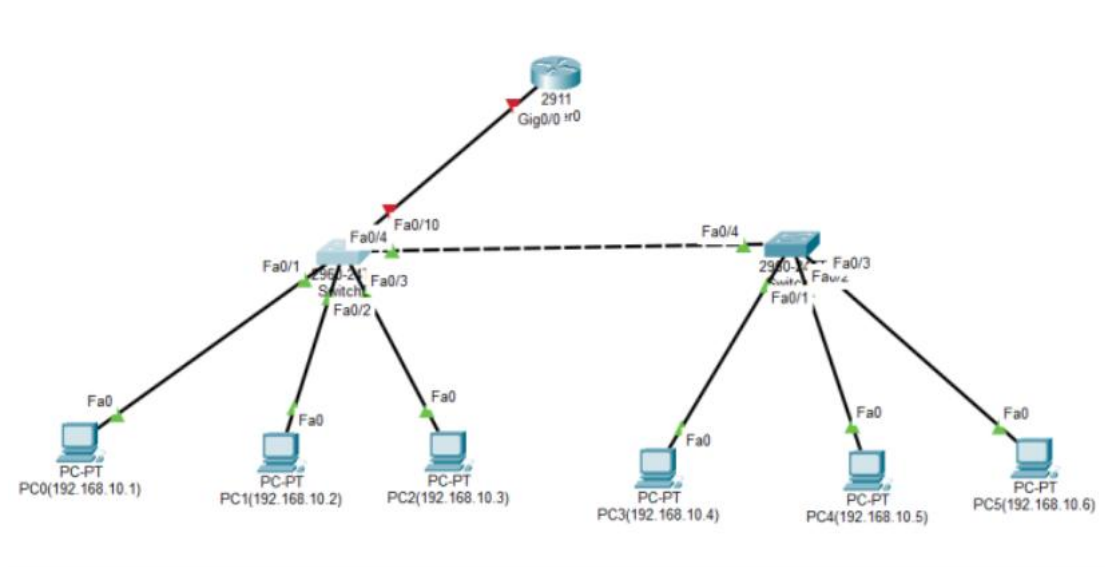
In real-world production environments, manually closing and reopening hundreds of ports can be extremely cumbersome. Cisco provides an automatic recovery function:

```
Switch(config)# errdisable recovery cause psecure-violation  
# Set the waiting time for automatic recovery (e.g. 30 seconds)  
Switch(config)# errdisable recovery interval 30
```

8. Experiment 4 — Single-Arm Routing Configuration

a. Upgrade Topology

Use a straight-through cable to connect Switch0 (Fa0/10) to Router 2911. (GigabitEthernet 0/0). Initially, both ends of the cable will appear red; this is normal because router interfaces are disabled by default.



b. Configure Single-Arm Routing

i. Switch Side: Backbone Link Configuration

To allow the link connecting to the router to carry tagged packets for multiple VLANs, the port must be set to Trunk mode. Enter the CLI for Switch0:

Switch>**enable**

#User Mode → Privileged Mode: Gain authorization to view and modify configurations.

Switch# **configure terminal**

#Privileged Mode → Global Configuration Mode: Prepare to modify switch parameters.

Switch(config)# **interface fastEthernet 0/10**

#Interface Configuration Mode: Specify the Fa0/10 port connected to the router.

Switch(config-if)# **switchport mode trunk**

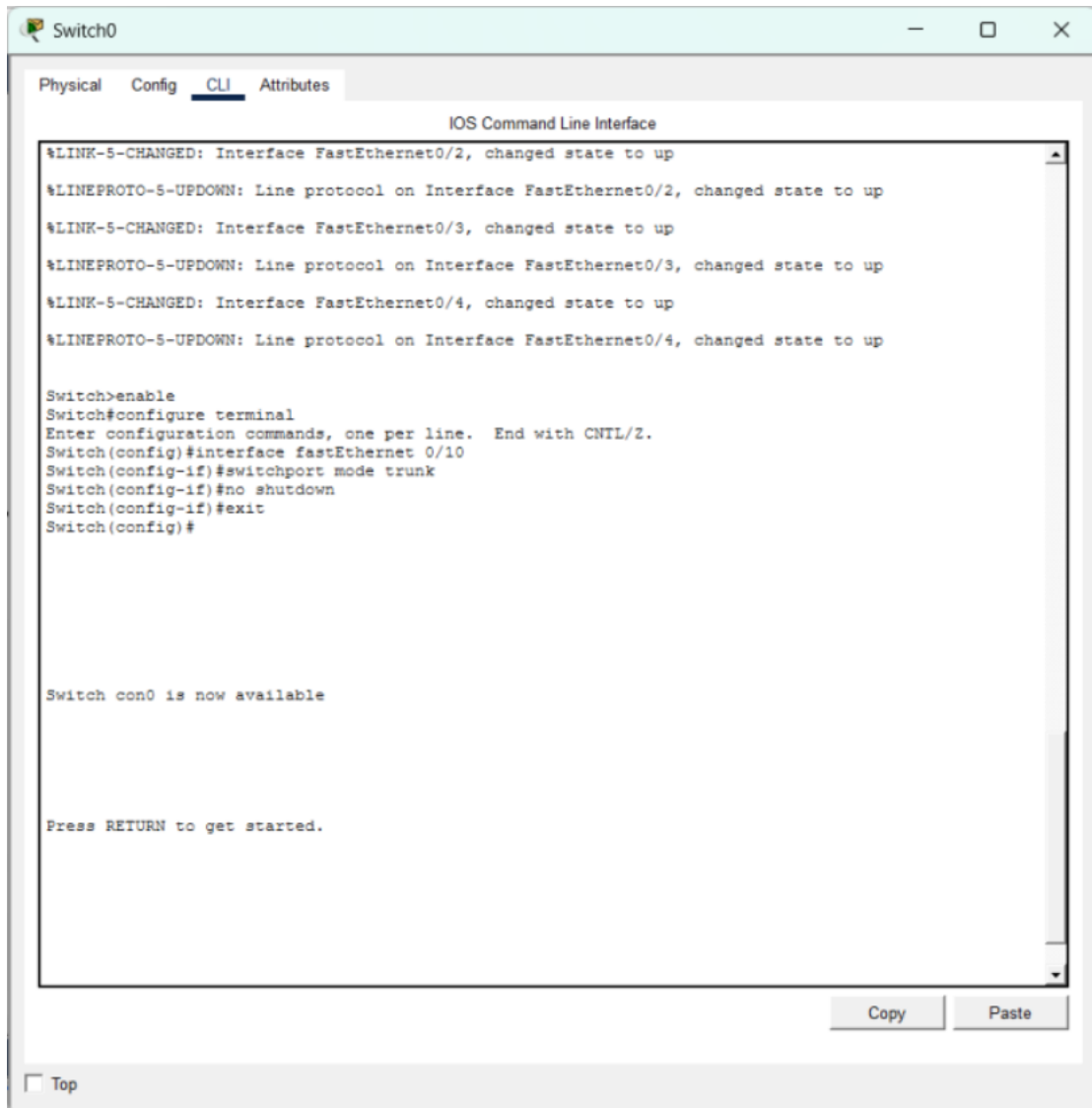
#Configure Trunking: Allow the link to carry data for multiple VLANs (10 and 20).

Switch(config-if)# **no shutdown**

#Activate Interface: Ensure the physical port is turned on.

Switch(config-if)# **exit**

#Exit Interface: Return to the previous Global Configuration level.



ii. Router Side: Sub-interface and Gateway Configuration

The router provides an independent Layer 3 entry point for each VLAN by dividing logical sub-interfaces. Enter the CLI for the Router:

```
Router> enable
```

Enter privileged mode: Ready to start configuring the router.

```
Router# configure terminal
```

Enter global configuration mode: Ready to enter configuration commands.

```
Router(config)# interface gigabitEthernet 0/0,
```

Physical interface mode: Enter the main physical interface to perform the enable operation.

Router(config-if)# **no shutdown**

Enable the physical master switch: The connection lines in the topology diagram will change from red to green at this point.

Router(config-if)# **exit**

Return to global configuration mode: Ready to create subinterfaces.

--- Subinterface 10 ---

Router(config)# **interface gigabitEthernet 0/0.10**

#Enter subinterface mode: Create a logical interface dedicated to VLAN 10.

Router(config-subif)# **encapsulation dot1Q 10**

Encapsulation protocol: Bind the 802.1Q protocol and specify that it belongs to VLAN 10.

Router(config-subif)# **ip address 192.168.10.254 255.255.255.0**

#Configure Gateway IP: Provide a routing exit for VLAN 10 hosts

Router(config-subif)# **exit**

#Return: Prepare to configure the next VLAN.

--- Sub-interface 20 ---

Router(config)# **interface gigabitEthernet 0/0.20,**

#Sub-interface Mode: Create a logical interface dedicated to VLAN 20.

Router(config-subif)# **encapsulation dot1Q 20,**

#Encapsulation Protocol: Bind the 802.1Q protocol to VLAN ID 20.

Router(config-subif)# **ip address 192.168.20.254 255.255.255.0,**

#Configure Gateway IP: Provide a routing exit for VLAN 20 hosts.

Router(config-subif)# **end**

#Finish Configuration: Exit back directly to Privileged Mode.

The screenshot shows the Router0 CLI interface with the following content:

```
Router0
Physical Config CLI Attributes
IOS Command Line Interface
Cisco CISCO2911/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
3 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

--- System Configuration Dialog ---
Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#interface gigabitEthernet 0/0.10
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up
encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.254 255.255.255.0
Router(config-subif)#exit
Router(config)#interface gigabitEthernet 0/0.20
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up
encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.254 255.255.255.0
Router(config-subif)#exit
Router(config)#
```

Buttons: Copy, Paste

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c. Verification Procedures

i. Terminal Configuration Verification

Click the PC icon, select the Desktop tab, and click IP Configuration to enter details:

PC0 (VLAN 10): IP 192.168.10.1, Default Gateway must point to 192.168.10.254.

PC3 (VLAN 20): IP 192.168.20.1, Default Gateway must point to 192.168.20.254.

PC0(192.168.10.1)

PhysicalConfigDesktopProgrammingAttributes

IP Configuration X

InterfaceFastEthernet0

IP Configuration

☐ DHCP

☒ Static

IPv4 Address

192.168.10.1

Subnet Mask

255.255.255.0

Default Gateway

192.168.10.254

DNS Server

0.0.0.0

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

/

Link Local Address

FE80::201:42FF:FE59:56D5

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

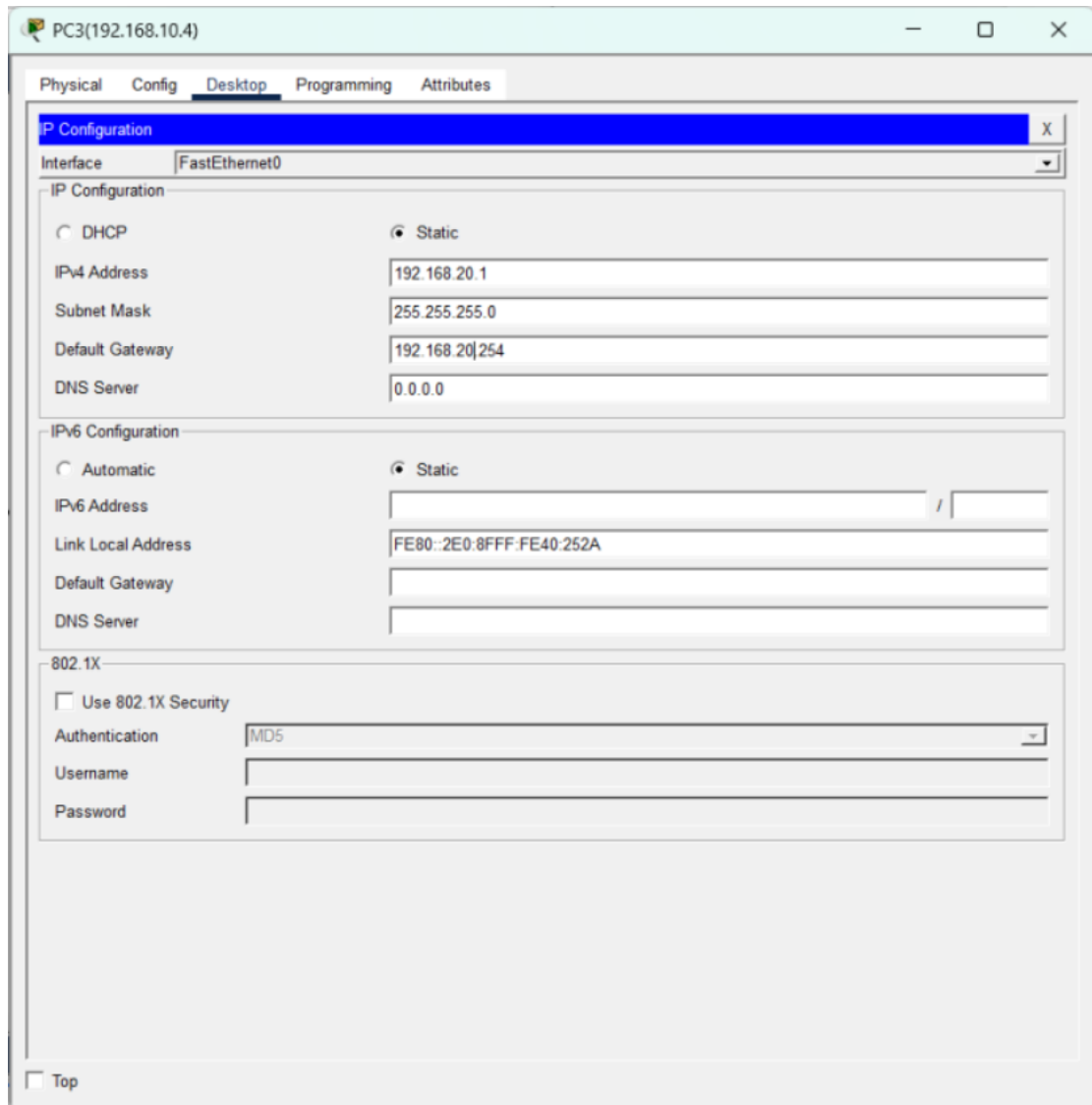
Authentication

MD5

Username

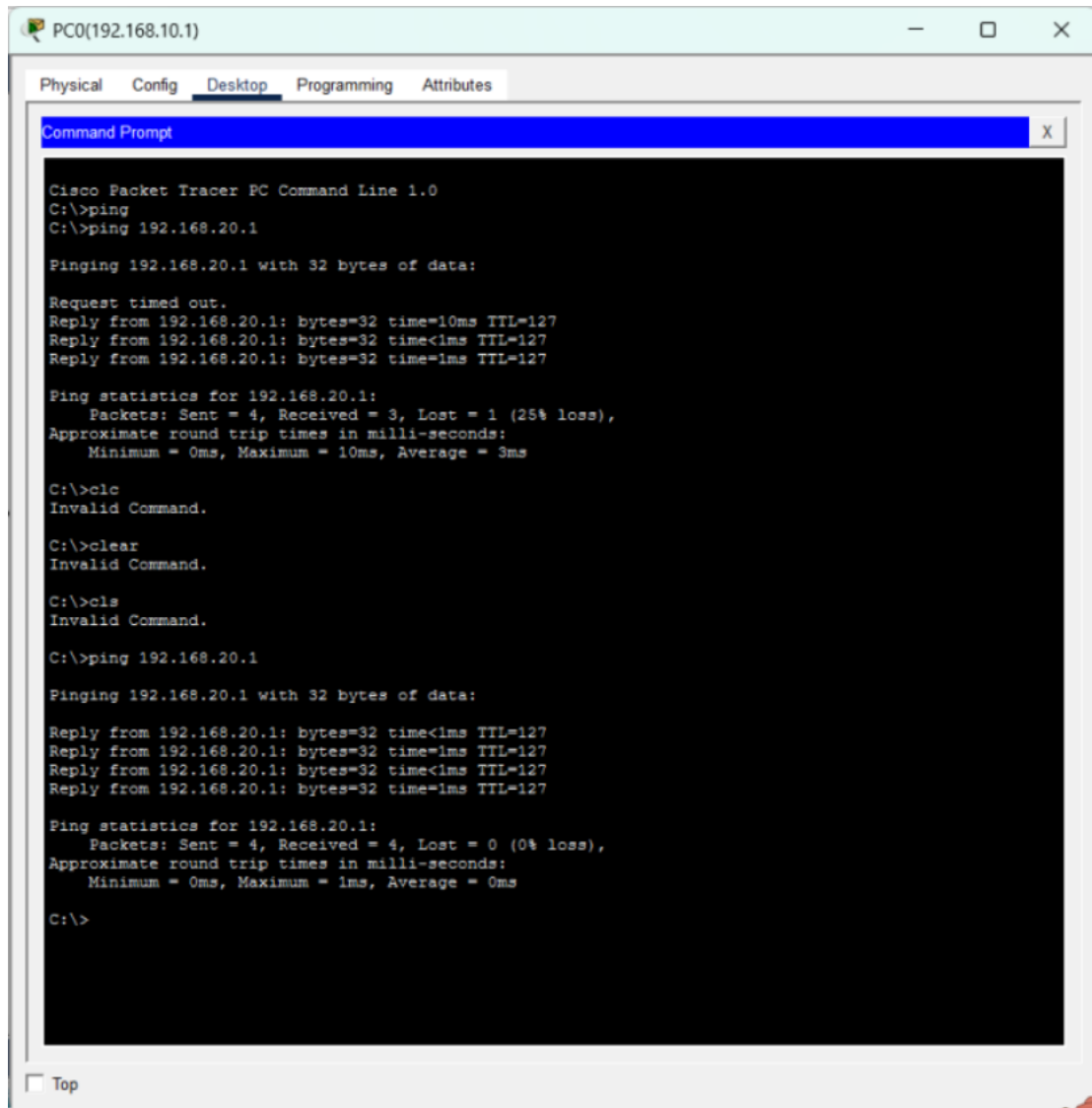
Password

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ii. Connectivity Test (Ping Test)

Open the Command Prompt on PC0 and enter ping 192.168.20.1. Due to ARP requests, the first 1-2 packets may "Timed out," followed by "Reply from...". Successful replies prove cross-segment communication is successful.



```
PC0(192.168.10.1)
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping
C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.1: bytes=32 time=10ms TTL=127
Reply from 192.168.20.1: bytes=32 time<1ms TTL=127
Reply from 192.168.20.1: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 3ms

C:\>clc
Invalid Command.

C:\>clear
Invalid Command.

C:\>cls
Invalid Command.

C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.20.1: bytes=32 time<1ms TTL=127
Reply from 192.168.20.1: bytes=32 time=1ms TTL=127
Reply from 192.168.20.1: bytes=32 time<1ms TTL=127
Reply from 192.168.20.1: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

iii. Routing Table Verification

Enter show ip route on the router CLI. You should see two routes marked with C (Connected) pointing to Gi0/0.10 and Gi0/0.20 respectively.

```
Router0
Physical Config CLI Attributes
IOS Command Line Interface
to comply with U.S. and local laws, return this product immediately.
A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wvl/export/crypto/tool/stqrg.html
If you require further assistance please contact us by sending email to
export@cisco.com.
Cisco CISCO2911/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
3 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)
Press RETURN to get started!
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up
Router>enable
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route
Gateway of last resort is not set
192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.10.0/24 is directly connected, GigabitEthernet0/0.10
L 192.168.10.254/32 is directly connected, GigabitEthernet0/0.10
192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.20.0/24 is directly connected, GigabitEthernet0/0.20
L 192.168.20.254/32 is directly connected, GigabitEthernet0/0.20
Router#
```

iv. Why is Trunk mode mandatory for the Switch-Router link?

v. What happens if the Default Gateway on the PC is configured incorrectly?

- vi. What are the limitations of the Single-Arm Routing approach in a high-traffic environment?
